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ASSIGNMENT = 3

[Github Link](#)



Numerical Computing in Python



NumPy (Numerical Python) – *Numerical Computing*

- **What it is:** A foundational library for numerical operations in Python.
- **Why it's useful:** It allows fast computation on large arrays and matrices.
- **Key features:**
 - Supports multidimensional arrays.
 - Provides mathematical functions like linear algebra, Fourier transforms, statistics.
- **Example:**

python

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```
import numpy as np
arr = np.array([1, 2, 3])
print(np.mean(arr)) # Output: 2.0
```



Pandas – Data Manipulation in Python



Pandas – *Data Manipulation*

- What it is: A library for handling and analyzing data structures, especially tabular data.
- Why it's useful: Great for working with data in CSV, Excel, SQL, etc.
- Key features:
 - DataFrame and Series objects.
 - Powerful group-by and pivot operations.
- Example:

python

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```
import pandas as pd
df = pd.DataFrame({'Name': ['Alice', 'Bob'], 'Age': [24, 27]})
print(df['Age'].mean()) # Output: 25.5
```



Matplotlib / Seaborn – Data Visualization



Matplotlib / Seaborn – Data Visualization

- What they are:
 - Matplotlib: A basic plotting library.
 - Seaborn: Built on Matplotlib, adds prettier and more complex visualizations.
- Why useful: Makes it easy to create charts and graphs for data analysis.
- Example (Seaborn):

python

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```
import seaborn as sns
import matplotlib.pyplot as plt
sns.set(style="darkgrid")
sns.barplot(x=['A', 'B', 'C'], y=[4, 7, 5])
plt.show()
```



Scikit-learn – Machine Learning



Scikit-learn – Machine Learning

- What it is: A powerful library for implementing machine learning algorithms.
- Why it's useful: Easy to use for classification, regression, clustering, etc.
- Key features:
 - Built-in models: decision trees, SVMs, k-NN, etc.
 - Preprocessing, cross-validation, model evaluation.
- Example:

python

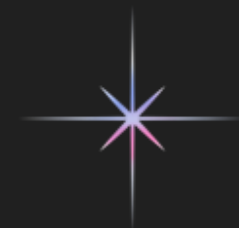
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```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit([[1], [2], [3]], [2, 4, 6])
print(model.predict([[4]])) # Output: [8.]
```



TensorFlow / PyTorch – Deep Learning



🧠 TensorFlow / PyTorch – Deep Learning

- What they are:
 - TensorFlow (by Google) and PyTorch (by Meta) are frameworks for building and training neural networks.
- Why they're useful: Used for deep learning tasks like image recognition, NLP, and AI applications.
- Key features:
 - GPU support for faster training.
 - Flexibility for building custom models.
- Example (PyTorch):

python

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```
import torch
x = torch.tensor([1.0, 2.0, 3.0], requires_grad=True)
y = x ** 2
y.backward(torch.ones_like(x))
print(x.grad) # Output: tensor([2., 4., 6.])
```



OpenCV – Computer Vision



🕒 OpenCV – Computer Vision

- What it is: A library for real-time image and video processing.
- Why it's useful: Can detect faces, objects, and perform image filtering.
- Key features:
 - Read, write, and manipulate images/videos.
 - Image transformation, edge detection, face/object tracking.
- Example:

python

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```
import cv2
image = cv2.imread('image.jpg')
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
cv2.imshow('Gray Image', gray)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



NLTK / spaCy – Natural Language Processing (NLP)



NLTK / spaCy – Natural Language Processing (NLP)

- What they are:
 - NLTK (Natural Language Toolkit): Ideal for teaching and research.
 - spaCy: Industrial-strength NLP library for faster performance.
- Why they're useful: Handle tasks like text classification, tokenization, and named entity recognition (NER).
- Example (spaCy):

python

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```
import spacy
nlp = spacy.load("en_core_web_sm")
doc = nlp("Dhruvi is working on a Python project.")
for ent in doc.ents:
    print(ent.text, ent.label_)
```



Pillow – Image Processing



Pillow – Image Processing

- **What it is:** A Python Imaging Library (PIL) fork used for opening, editing, and saving images.
- **Why it's useful:** Ideal for basic image processing tasks like resizing, cropping, filtering, adding text, etc.
- **Key features:**
 - Supports many image formats (JPEG, PNG, BMP, GIF, etc.)
 - Easy to manipulate pixels and apply effects
 - Can generate thumbnails, convert formats, and overlay images
- **Example:**

```
python                                                                    Copy Edit

from PIL import Image

# Open an image
img = Image.open("example.jpg")

# Resize the image
img_resized = img.resize((100, 100))

# Save the image
img_resized.save("resized.jpg")
```



Keras – High-Level Deep Learning API



🧠 Keras – High-Level Deep Learning API

- **What it is:** A high-level neural networks API, written in Python and capable of running on top of TensorFlow.
- **Why it's useful:** Makes building and training deep learning models easy and beginner-friendly.
- **Key features:**
 - Simple and intuitive syntax.
 - Pre-trained models for image and text tasks.
 - Works well for fast prototyping.
- **Example:**

```
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from keras.models import Sequential  
from keras.layers import Dense  
  
# Create a simple neural network  
model = Sequential([  
    Dense(32, activation='relu', input_shape=(10,)),  
    Dense(1, activation='sigmoid')  
)  
  
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```



TensorFlow – Deep Learning Framework



● TensorFlow – Deep Learning Framework

- What it is: An open-source library developed by Google for machine learning and deep learning.
- Why it's useful:
 - Highly flexible and powerful.
 - Supports both low-level operations and high-level APIs (like Keras).
 - Used in real-world AI applications: image recognition, NLP, etc.
- Fun fact: TensorFlow supports GPU acceleration, making training much faster.

✓ Short Example: Linear Regression with TensorFlow

```
python                                                                    Copy Edit

import tensorflow as tf
import numpy as np

# Sample data
X = np.array([0, 1, 2, 3, 4], dtype=float)
Y = np.array([0, 2, 4, 6, 8], dtype=float) # y = 2x

# Build the model
model = tf.keras.Sequential([
    tf.keras.layers.Dense(units=1, input_shape=[1])
])

# Compile the model
model.compile(optimizer='sgd', loss='mean_squared_error')

# Train the model
model.fit(X, Y, epochs=100, verbose=0)

# Predict a new value
print(model.predict([5])) # Expected output: ~10
```





THANK YOU!